

The empirical recurrence for OEIS A235959: recovering a conjecture that is not written down

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Abstract

OEIS A235959 records “Empirical recurrence of order 67”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 181$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 67, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 67 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235959 is “Number of $(n+1) \times (5+1)$ 0..1 arrays with the difference of the upper and lower median value of each 2×2 subblock in lexicographically nondecreasing order columnwise and nonincreasing rowwise.”. It has offset 1 and begins

1076, 13346, 137988, 1340034, 12412248, 112308598, 997114552, 8749889970, ...

The entry states:

Empirical recurrence of order 67 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:57:13, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 2113$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 181$ of the 2113, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 181$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 4226$ exact terms of the model returns order 67 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{67} c_i a(n-i),$$

c_1	15	c_{18}	−90314885054167	c_{35}	70055842972370052754	c_{52}	−458069126898772
c_2	105	c_{19}	−57228743002022	c_{36}	89596079516024084676	c_{53}	−6030336832008911
c_3	−2465	c_{20}	884393365110078	c_{37}	−182616803308599845056	c_{54}	4827051764266670
c_4	−2162	c_{21}	119807827883597	c_{38}	−169192927255497515456	c_{55}	28703202833358179
c_5	183925	c_{22}	−7074928823523671	c_{39}	398829087763623394872	c_{56}	−282572405563163
c_6	−235784	c_{23}	1792413174201672	c_{40}	263838015613420814000	c_{57}	−1070476048977037
c_7	−8293502	c_{24}	46709616532311180	c_{41}	−731132898802366038928	c_{58}	1085981367967573
c_8	20812314	c_{25}	−26660529873510378	c_{42}	−334841762235224617664	c_{59}	3034738097019630
c_9	253322822	c_{26}	−256378521637775124	c_{43}	1124714155585560358656	c_{60}	−273849684901389
c_{10}	−870912926	c_{27}	215769597661431640	c_{44}	337012754662571923968	c_{61}	−624574549629232
c_{11}	−5555914513	c_{28}	1175874539207831785	c_{45}	−1448683322116316613632	c_{62}	417178213639258
c_{12}	24010984291	c_{29}	−1274132868220415204	c_{46}	−255050587758063826944	c_{63}	864587774362124
c_{13}	90174986894	c_{30}	−4520942861116851971	c_{47}	1556082439433080442880	c_{64}	−30342122880106
c_{14}	−482512637888	c_{31}	5925713375376412061	c_{48}	124886288046366846976	c_{65}	−69559503619620
c_{15}	−1092705025417	c_{32}	14591521493657458495	c_{49}	−1385453984783980867584	c_{66}	28147497671063
c_{16}	7429237718840	c_{33}	−22411129555920400297	c_{50}	−10474199952177348608	c_{67}	22517998136852
c_{17}	9692075008890	c_{34}	−39518969816462383334	c_{51}	1013924403781651300352		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 67, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $67 + 4$ terms removed returns order 67 as well, so $\deg q_{\min} = 67 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $67 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 67 that this sequence satisfies — and that family turns out to have exactly one member.

References

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